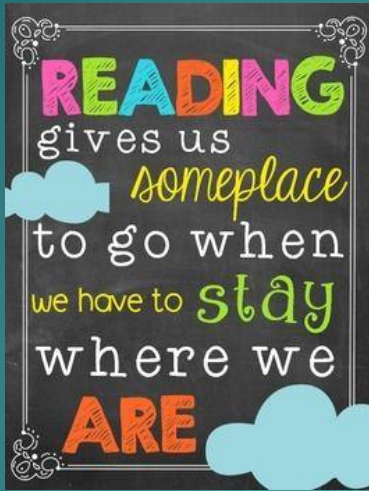


Lecture 1

Web User Experience & Database Design

By Joop
Inthraporn Aranyanak, PhD





Inthraporn Aranyanak, Phd

“Joop”

EXPERTISE: UX research, UI design, Web development, Design thinking, UX coaching

WORK

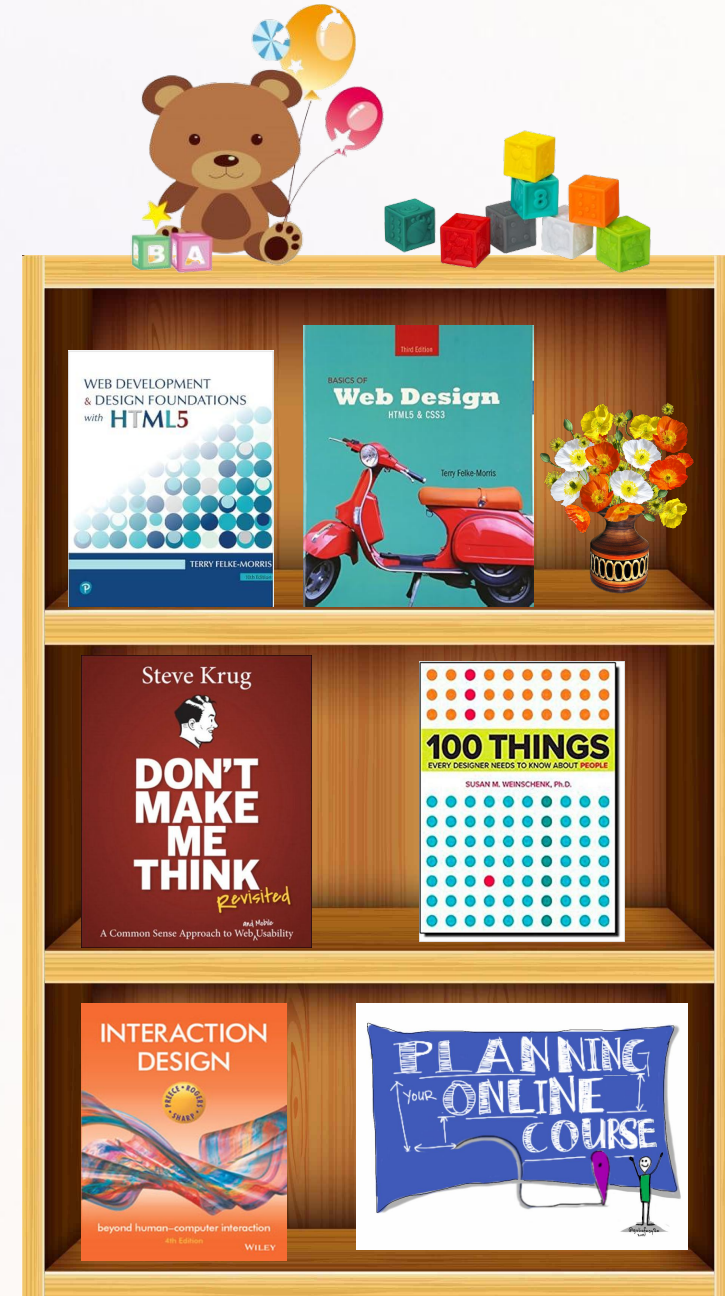




Week 1 - Greeting



- 
- Course Description
 - The Goal of Module
 - Module Evaluation
 - Module Schedule
 - Software Development Process
 - Intro to Human & Computer Interaction





More Resources



Interaction-Design.org

394K like this · Non-profit organisation
Learn about UX Design through our open-source educational mate...

☒ Liked ▼



UI/UX designers

84K members · 10+ posts a day

This group is meant for designers - a place to learn and share - to ask questions, network, and improve....



HTML5 / CSS3 / Javascript (HTML / CSS / JS)

102K members

Discussions on **HTML5**, CSS3 and new Javascript. New innovative ways of developing universal web/app. This Group is for experts profession...



Figma Product Community

185K members

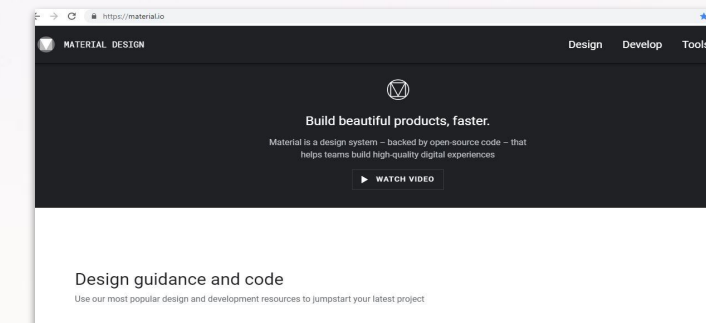
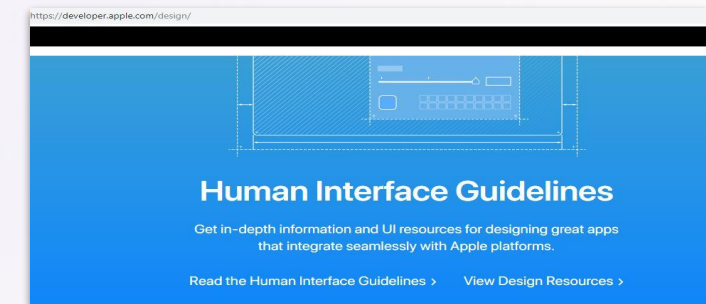
A community for **Figma** users looking to learn from others, share tips and tricks and expand our skillset. Your go to place for resources and...



Figma Designers

8K members

Sharing and discussing latest and coolest stuff about **Figma**





Course Description

This module introduces the foundational principles of user interface (UI) and user experience (UX) design, emphasising the development of intuitive, user-friendly digital interfaces. Students will gain theoretical knowledge and practical skills in website construction and design, informed by industry best practices.

The module also covers essential database concepts, including relational database design, integrity, and referential constraints. Students will also be introduced to the concept of NoSQL Databases.

Students will integrate these skills to build functional websites connected to databases, capable of performing essential queries such as insertions, updates, and selections. This hands-on approach equips students with the ability to design, plan, and implement cohesive web solutions.



Draft! - Module Assessment Semester 1 & 2 (100%)

Semester 1

- Quiz (Week 13) – 10% (WebUX 5% + Database 5%)
- Lab Activities – 10% (WebUX 5% + Database 5%)
- End-of-lecture questions – 10% (WebUX 5% + Database 5%)
- Individual Project – 20%
- CA1: WebUX – 10%
- CA2: Database Design – 10%

Semester 2

- Quiz - TBC 10%
- Lab Activities 10%
- Team Project (Focus on Web Dev & Database Design) 30%

Note: Attendance will be recorded at each class session.

It might be updated later.

TBC = To be confirmed





Lab Assessment for Web UX

- **You must present your work to the lecturer in person before leaving the lab.**
- Then submit your work on Brightspace by the due date provided on Brightspace.
- Work not presented in person will not be graded, even if it is submitted on Brightspace.





Tools used in this module



Design and Prototyping:

Miro: For design collaboration, flow diagrams, and creating Mockups

Development:

HTML: For structuring the content of web pages.

CSS: For styling and layout.

JavaScript: For adding interactivity. (Optional)



Module WebUX Schedule

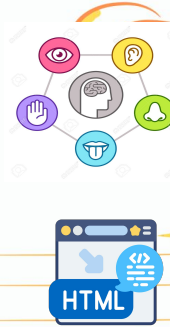
Topic 1

- Module Intro & Intro to HCI
- Software Development process



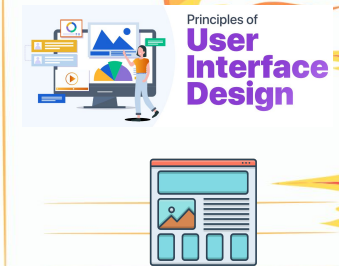
Topic 2

- Human Perception
- Intro to HTML



Topic 3

- UI Principles
- UI Design



Topic 4

- Semantic HTML
- Intro to CSS



Topic 5

- Inclusive Design
- CSS (Box Model)



Please note: it might be updated later.

Module Database Design Schedule

Topic 1

Information, Data, and
Persistence

Topic 2

Asking and Searching
Data with SQL

Topic 3

Adding, Correcting, and
Protecting Information

Topic 4

When One Table Is
Not Enough

Topic 5

Designing a Small
Information Structure

Please note: it might be updated later.





CrossList-Web User
Exp Data Design-
WADT-H1002-Sept-
Dec Academic Term
2026/27-4

My Home Progress ▾ Content Assessment ▾ Module Tools ▾ Library Help ▾

[Print](#) [Settings](#)

General Information ▾

Add dates and restrictions... 👁️

Welcome to WADT 1002: Web User Experience & Database Design

Module Overview

This module introduces **the foundational principles of user interface (UI) and user experience (UX) design**, emphasising the development of intuitive, user-friendly digital interfaces. Students will **gain theoretical knowledge and practical skills in website construction and design**, informed by industry best practices.

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Module Assessment

Semester 1

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- Lab Activities – 10% (WebUX 5% + Database 5%)
- End-of-lecture questions – 10% (WebUX 5% + Database 5%)
- Individual Project – 20%



**What do we need to know to build
web or mobile apps?**

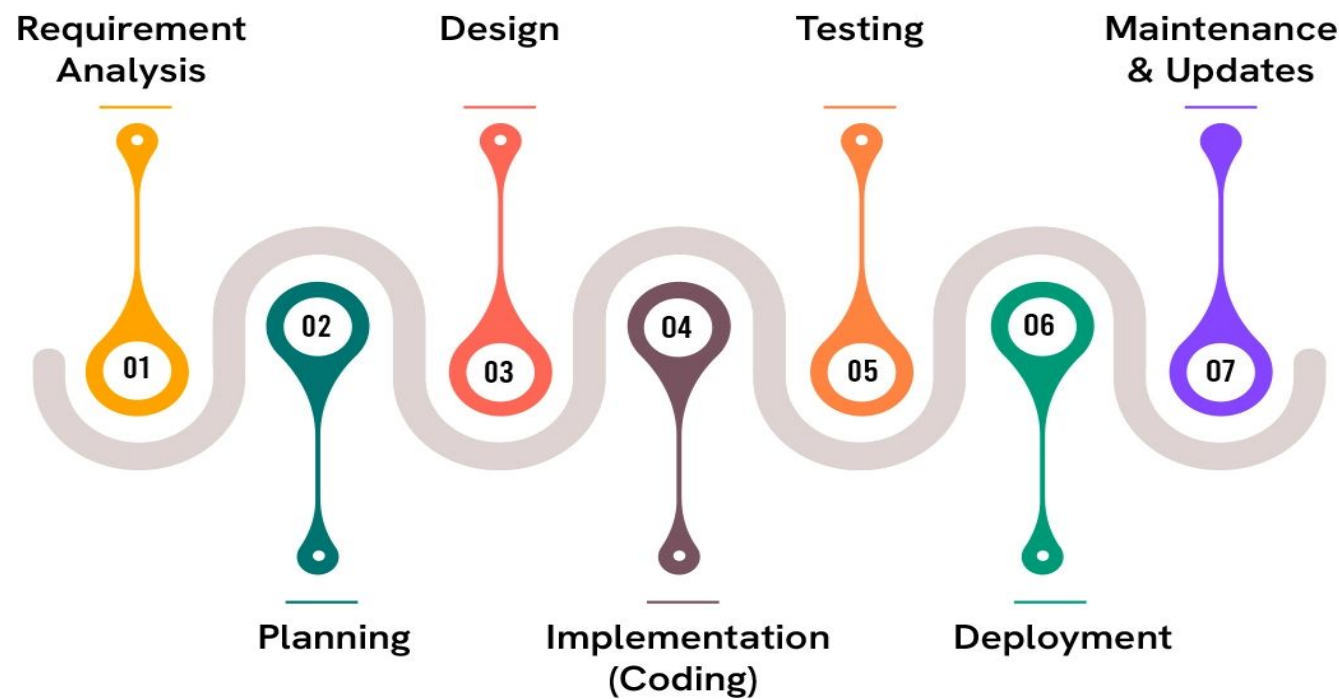




Software Development Process

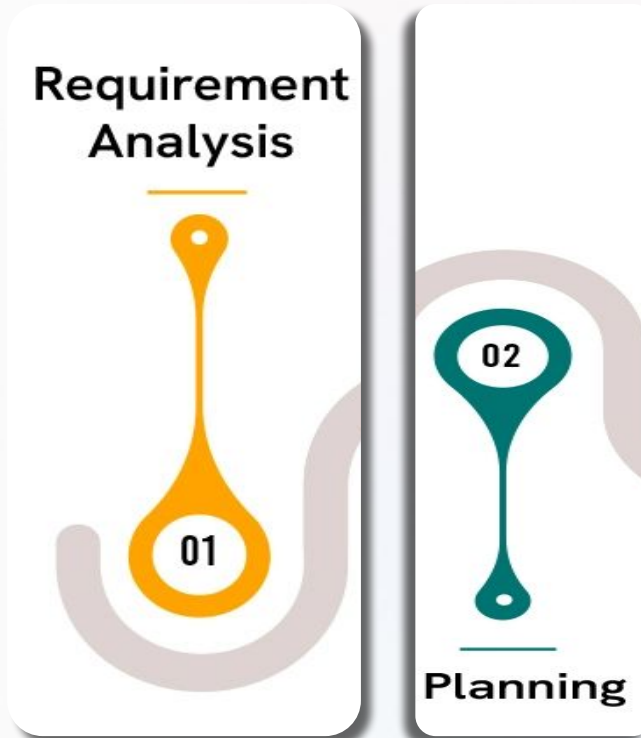


STEP-BY-STEP: HOW THE SOFTWARE DEVELOPMENT PROCESS WORKS





Software Development Process

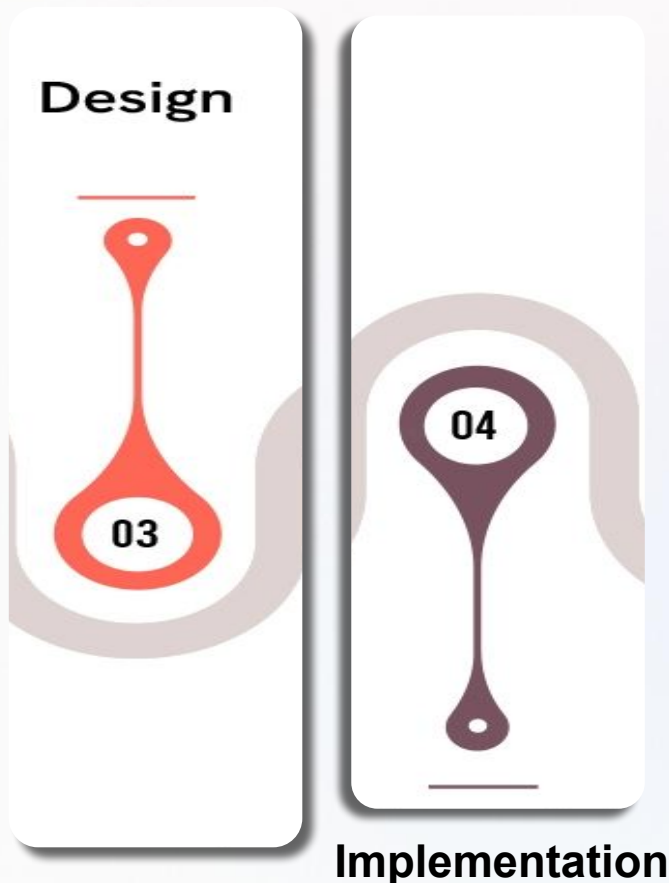


1. **Requirement Analysis:** is the first and essential phase of software development. It involves identifying, understanding, and documenting stakeholder needs to define what the software must achieve. This includes: Functional requirements – specific features or behaviors (e.g., user login, data entry) and Non-functional requirements – performance, security, usability, scalability, etc.
2. **Planning:** The planning phase lays the foundation for a successful software project by defining goals, objectives, and requirements based on customer feedback or market research. Stakeholders collaborate to establish the project scope, timelines, and resource allocation, ensuring a clear direction and shared understanding among all participants.





Software Development Process

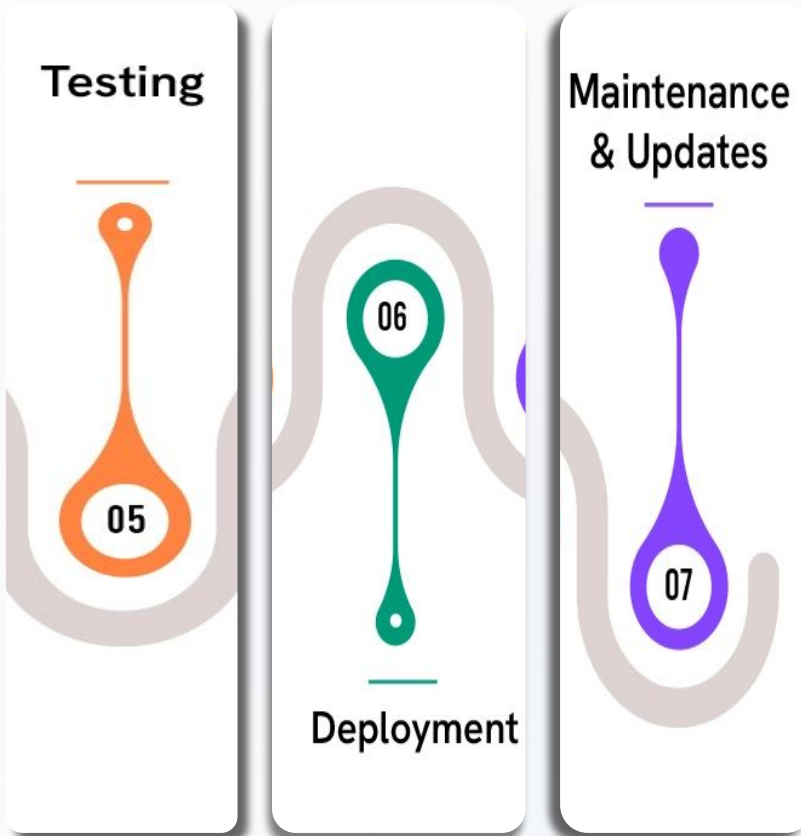


3. **Design:** The system design phase defines the software's architecture and functionality based on gathered requirements. It includes high-level architecture, detailed design specifications, user interface design, and compatibility assessment to ensure usability and integration with existing products.
4. **Implementation:** The implementation phase converts the design into a functional application through coding. Developers follow design specifications, best practices, and coding standards to ensure the software is efficient, secure, and maintainable.





Software Development Process



5. **Testing:** This stage verifies that the software meets all requirements and functions correctly. It involves identifying and fixing bugs, checking performance, and ensuring reliability through various tests (e.g., user, unit, system, UAT, performance, security, etc.).
6. **Deployment:** Once tested and approved, the software is released to the production environment – known as the Go Live stage. This may involve installation, configuration, user training, and monitoring to ensure a smooth transition and operation for end-users.
7. **Maintenance & Updates:** The final phase ensures the software remains functional and up-to-date after deployment. It involves fixing issues, applying updates, and adding new features to keep the system reliable and relevant.





Scrum Framework





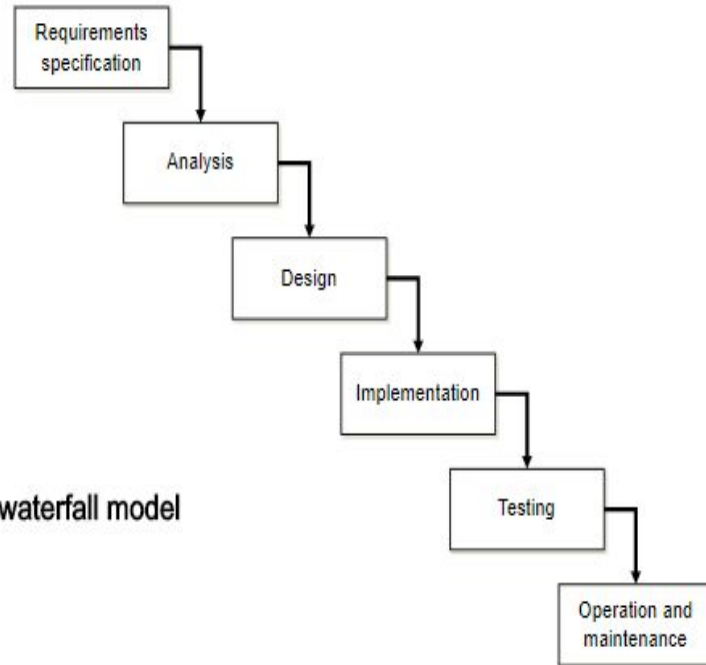
Traditional approach in Software development process

Advantages:

1. Simple & Easy to Manage – The step-by-step structure makes it straightforward to follow and manage.
2. Clear Documentation – Each phase is well-documented, making it easier for teams to understand and follow.
3. Good for Stable Projects – Works well when requirements are clear, fixed, and unlikely to change.

Disadvantages:

1. Hard to Make Changes – Once a phase is complete, going back to modify things is difficult and costly.
2. Slow Development Process – Each phase must be fully completed before moving to the next, which can slow progress.
3. Late Discovery of Issues – Bugs and issues are often found late in the process, making fixes expensive.



The waterfall model

<http://slideplayer.com/slide/7431292/>





Agile approach in Software development process

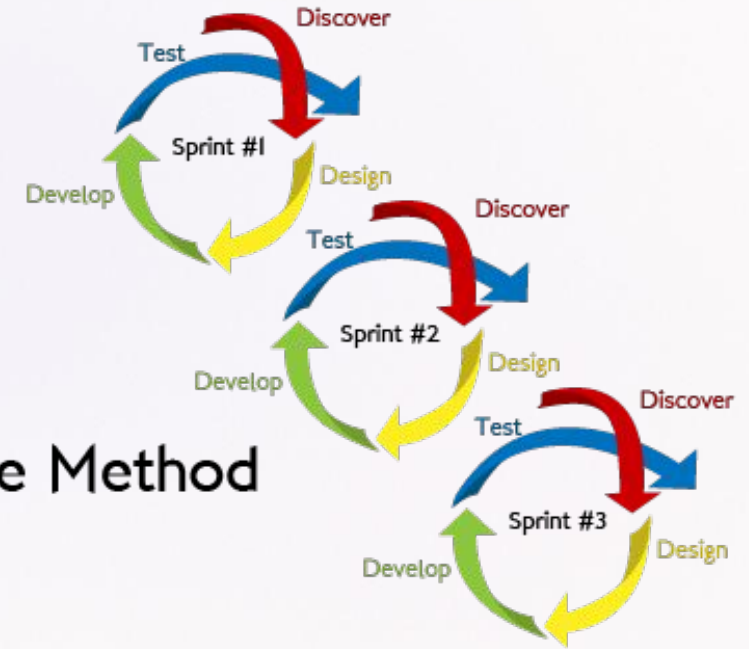
Advantages:

1. More Flexibility – Easily adapt to changing requirements.
2. Faster Time-to-Market – Delivers working software quickly in smaller increments.
3. Better Collaboration – Encourages regular feedback from customers and stakeholders.

Disadvantages:

1. Less Predictability – Harder to estimate time and cost upfront.
2. Requires Active Client Involvement – Success depends on continuous customer feedback.
3. Can Be Challenging for Large Projects – Managing Agile for big teams may require extra coordination.

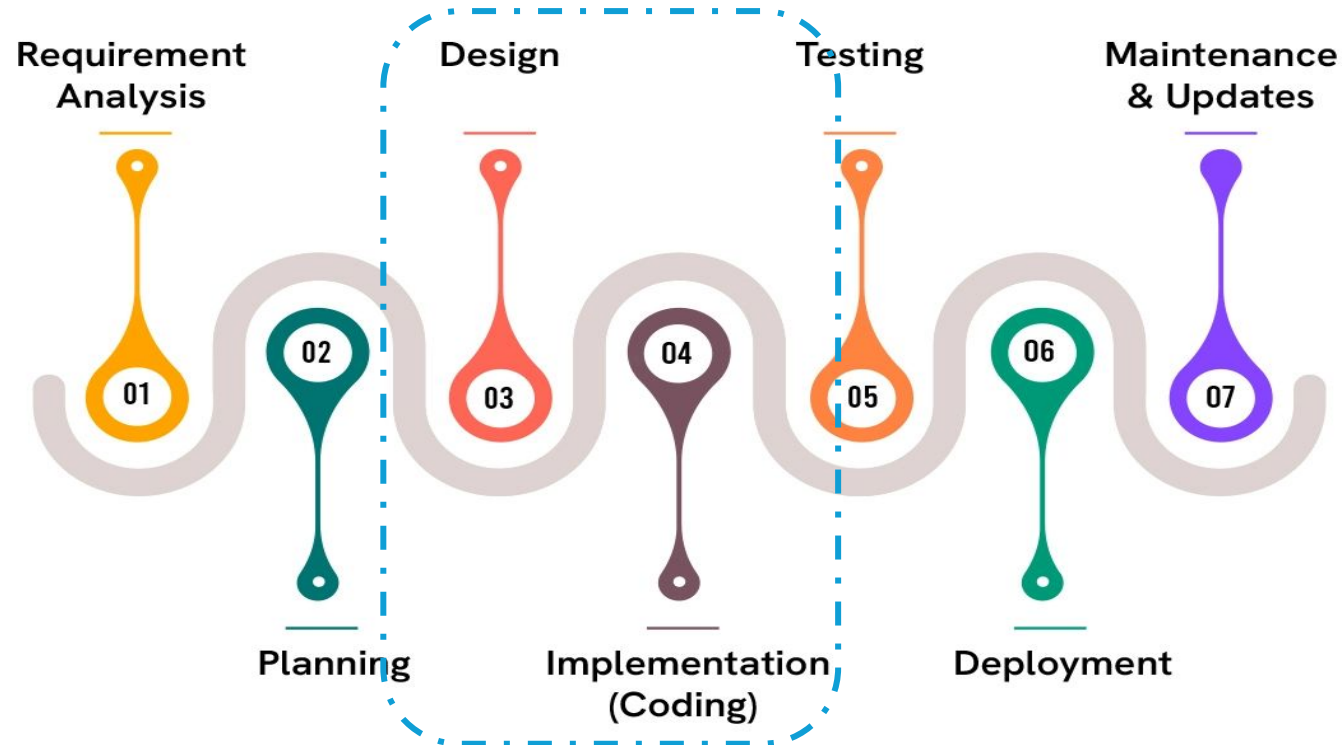
Agile Method



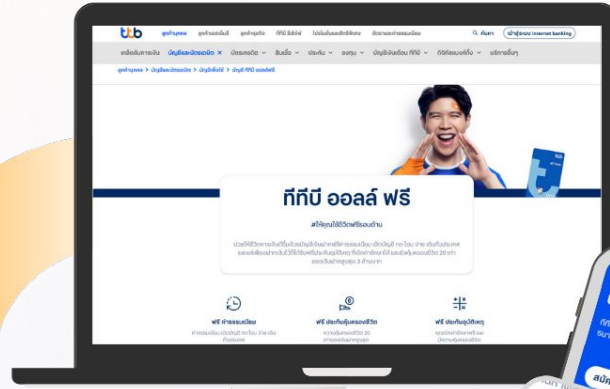
This module focuses on:



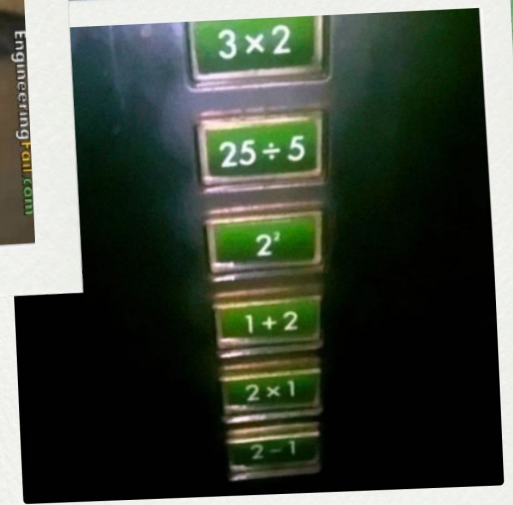
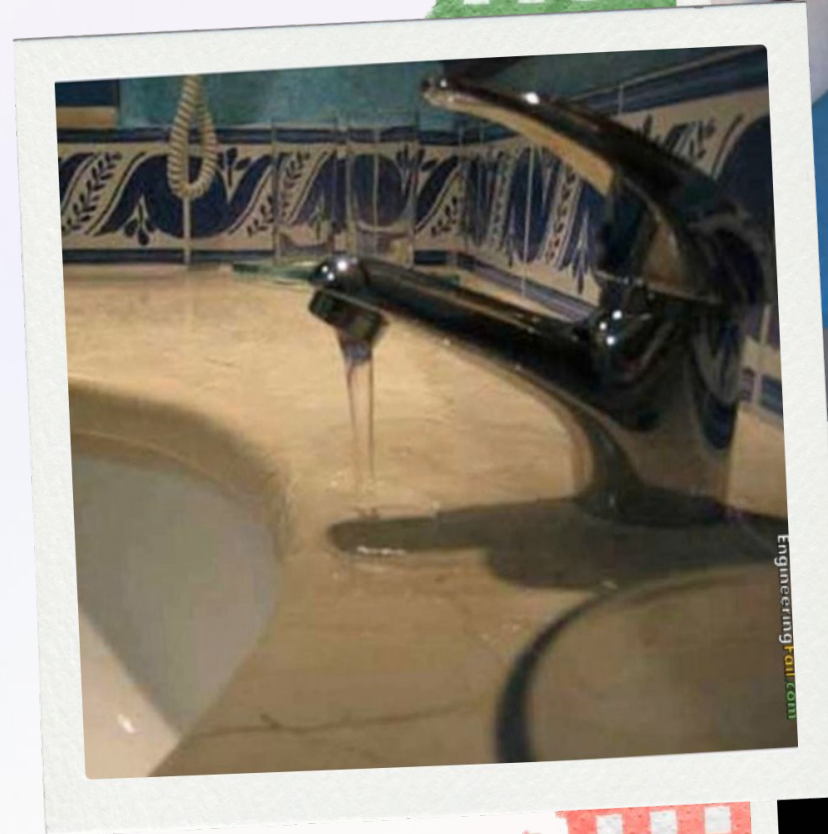
STEP-BY-STEP: HOW THE SOFTWARE DEVELOPMENT PROCESS WORKS



Design



Have you ever seen these?





What should you consider when designing?



Some examples of UI Design Based on UX

The Evolution of Netflix UI/UX

From DVD rentals to a personalized streaming experience

Same goal, better experience.

1997 – 1999
DVD Rental by Mail

- Simple, text-based design
- Focus on DVD rentals by mail
- Users browsed movie lists and added them to a queue

Goal: Make movie rental convenient

2000s
Early Streaming

- Transition to online streaming
- Grid-based layout
- Movie covers, ratings, and categories

Goal: Bring movies online

2010
A More Visual Experience

- Larger, richer visuals
- Personalized rows (e.g., Continue Watching)
- Smoother streaming experience across devices

Goal: Make discovery easier

2017
A Personalized Home

- Bigger hero banner
- Personalized recommendations based on viewing behavior
- Clear actions (Play, My List)
- Works seamlessly across devices (TV, mobile, web)

Goal: Personalize and engage

2023 – Today
Smarter, More Immersive

- Richer visuals and motion (trailers, autoplay)
- Smarter, more precise recommendations using AI
- Personalized rows (e.g., Top 10 in your country)
- Consistent, high-quality experience across all devices

Goal: Keep you watching longer

Key UX/UI Improvements Over Time

From static to visual
Evolved from text lists to immersive, image-rich design.

From one-size-fits-all to personalized
Recommends content based on your behavior and preferences.

From single device to multi-device
Seamless experience across web, mobile, tablet, and TV.

From browsing to instant play
Autoplay, previews, and one-click play reduce friction.

From finding content to keeping you engaged
Personalized rows, Top 10 lists, and dynamic artwork create curiosity and habit.

Same mission. A smarter experience.

Netflix has continuously evolved its UI/UX by combining technology and psychology to make entertainment more accessible, personal, and engaging.

Because our biggest competitor is sleep.

Original (No persuasive messages)

New Version (With persuasive messages)

Types of Persuasive Messages

- Scarcity**
"Only 3 rooms left at this price!"
Creates a fear of missing out (FOMO).
- Social Proof**
"18 people are looking at this property right now"
Shows that others are interested.
- Urgency**
"27 bookings in the last 24 hours"
Signals high demand and encourages immediate action.

Expected Results
From persuasive messages

- Higher Click-Through Rate (CTR)
- Higher Conversion Rate
- Shorter Decision Time

This study examines how persuasive messages on Booking.com may influence users' booking decisions.



**“Design is not just about aesthetics;
it must also take user experience into
account.”**



Keep in Mind When Designing



Purpose

What is the goal of this design? What problem does it solve?



Audience

Who is it for? What are their needs, preferences, or limitations?



User Experience (UX)

Is it easy and intuitive to use? Does it feel smooth and user-friendly for the user?



Aesthetics

Is it visually appealing? Are the colors, fonts, and layout harmonious?



Clarity

Is the message clear? Is there too much or too little information?



Consistency

Are visual and interactive elements used consistently throughout the design?



Accessibility

Can everyone – including people with disabilities – access and understand easily?





“ To design better experiences, we must understand the way humans interact with computers. This broader perspective is what we call Human-Computer Interaction (HCI). ”





Checking your attendance



End of Part 1



WEB UX DESIGN



An Introduction to HCI

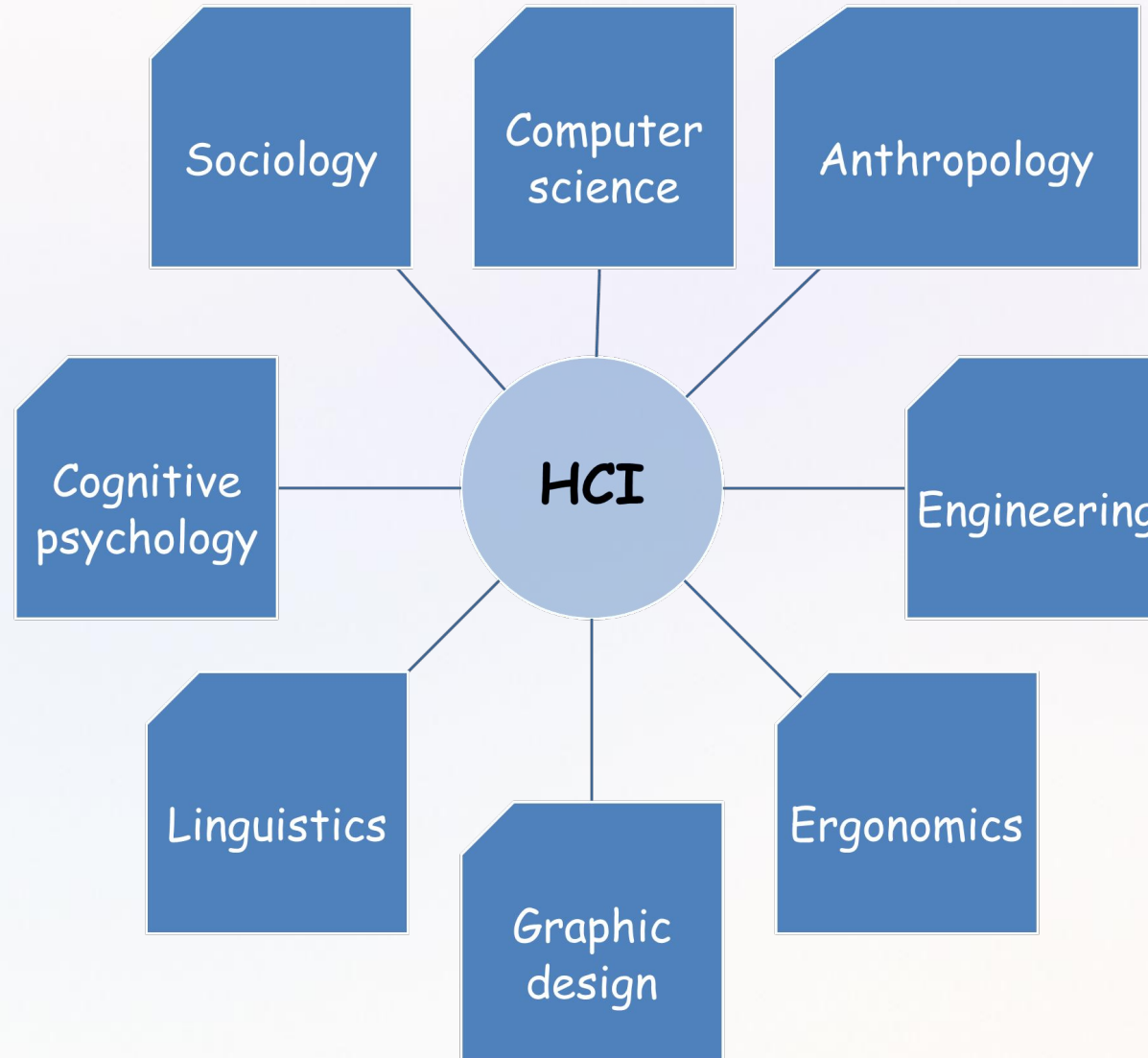
"Human-computer interaction is a discipline concerned with the design, evaluation, and implementation of interactive computing systems for human use and with the study of major phenomena surrounding them."

- **Thomas T. Hewett et al., 1992**





Human-computer interaction and related fields





The history of HCI

1950s – Computers Invented

1960s – first screen and mouse developed

1970s - Robots

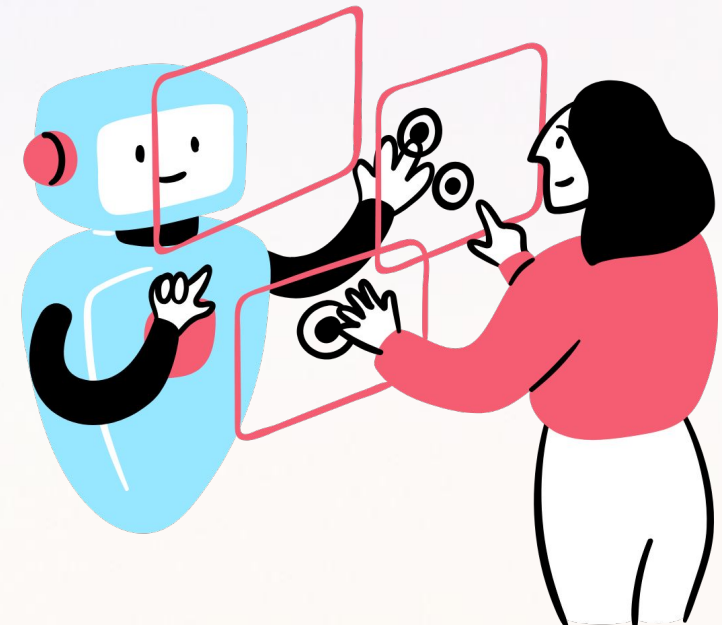
1980s – Arrival of microchip and micro-computers

1984 – Apple Macintosh (Xerox Star). First conference in HCI.

1990s – World Wide Web arrives: Websites & Smartphones

2000s - AR/VR technologies

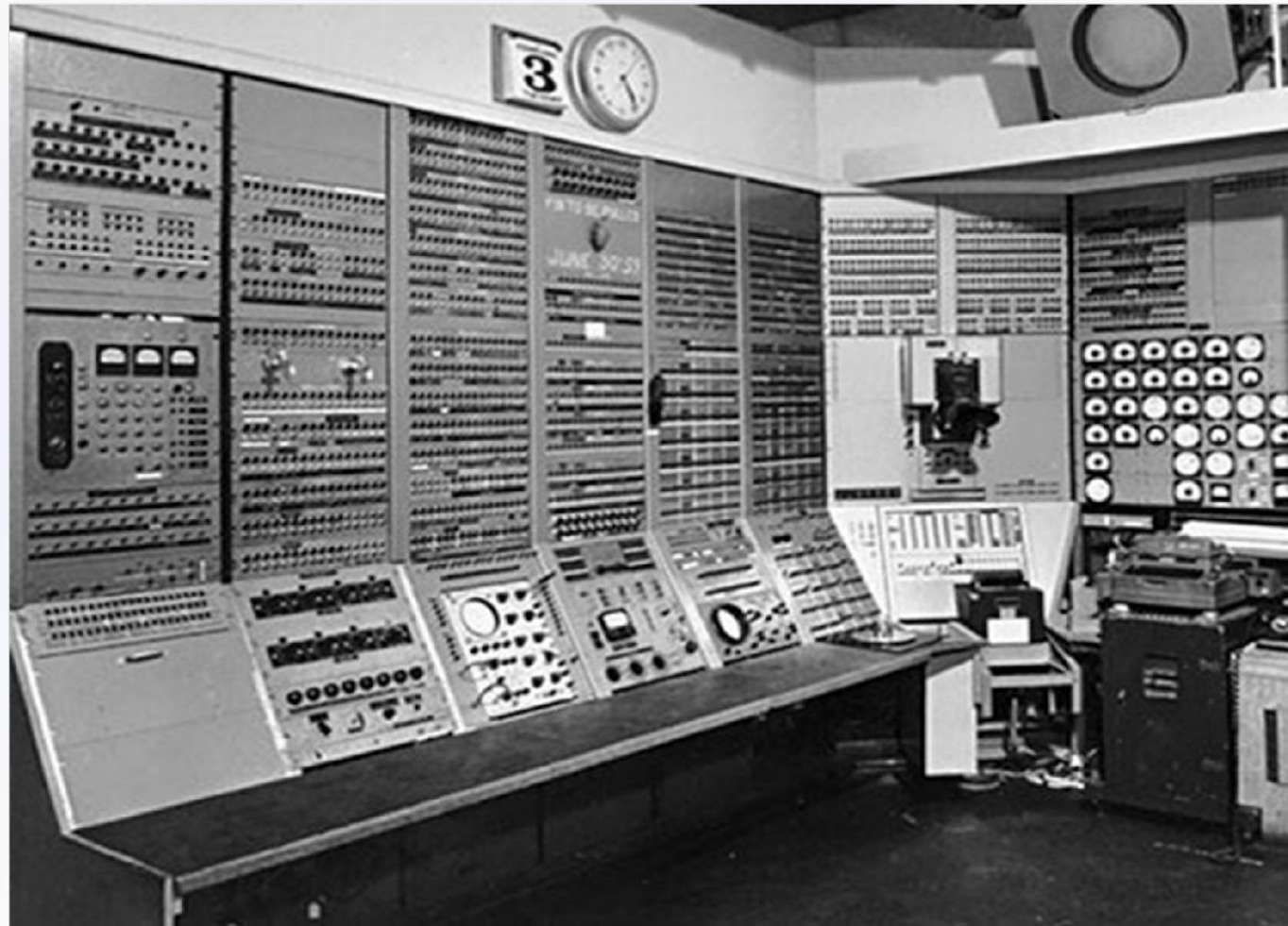
Present - Trends in AI-Driven Human-Computer Interaction





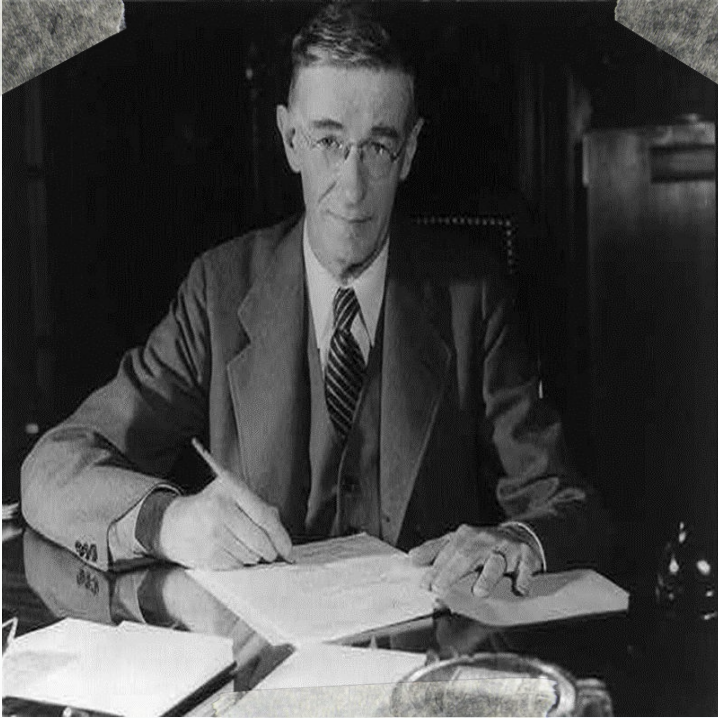
First Computer

First computer with RAM - 1955





Vannevar Bush

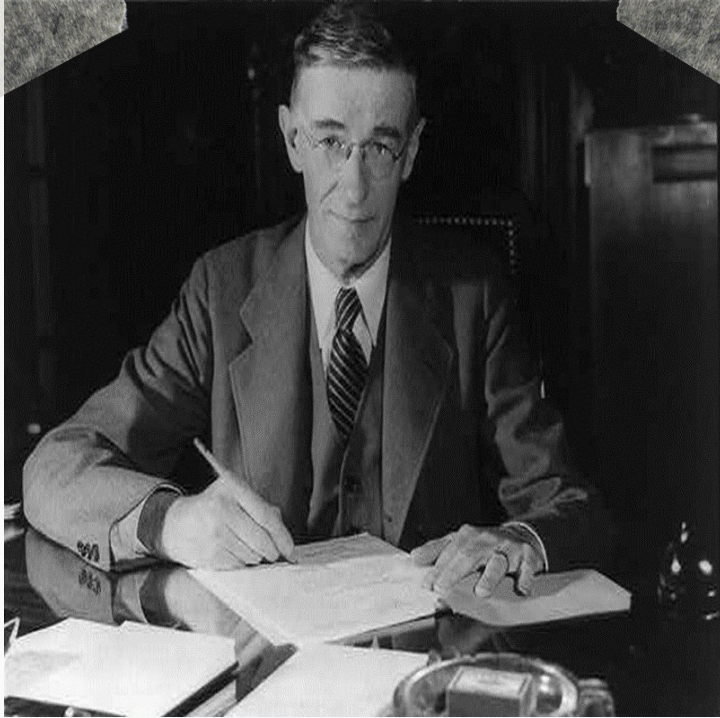


- He was an American engineer, inventor and science administrator
- He published "As We May Think" - 1945 Atlantic Monthly
- He Developed Memex
 - Stores all records/articles/communications
 - Items retrieved by indexing, keywords, cross references (now called hyperlinks)
 - (Envisioned as microfilm, not computer)
 - So, you may think of it like an early version of hyperlinks.
 - It helped inspire the internet and the World Wide Web.





Vannevar Bush



https://www.youtube.com/watch?v=pW4SS_9nXyo





Ivan Sutherland



- He is known as the father of computer graphics using SketchPad .
- Sketchpad was a computer program written by Ivan Sutherland in 1963 in the course of his PhD thesis.
- Sketchpad was a computer program that let users draw and interact with graphics using a light pen.
- Instead of only writing code, people could draw, click, and interact directly on the computer screen through visual interfaces.
- It introduced many ideas that are now fundamental to modern computer graphics and human-computer interaction (HCI)





Ivan Sutherland





Douglas Engelbart



Guess what's in his hand?





Douglas Engelbart





Xerox Star



- It is the first personal computer using for “business professionals”
- It provides the concept of **WYSIWYG**.
- This means the content you see on the screen looks the same as the final output, like a printed document.
- Developed GUI for easy to use.
- The cost was about \$75,000

“What You See Is What You Get”





Apple Macintosh

- Developed a better GUI to be easy to use
- User interface friendly
- It is a personal computer
- The cost was about \$2,500



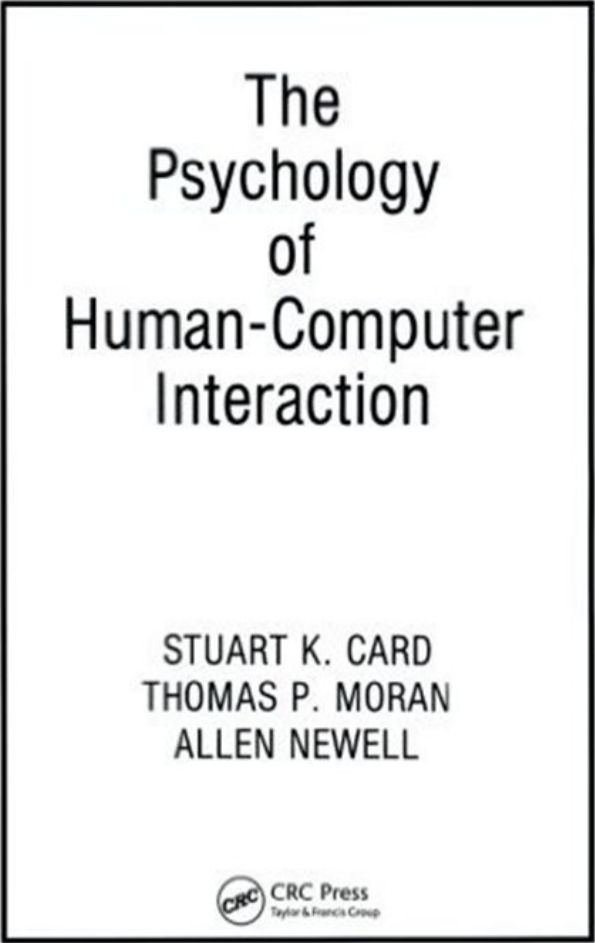


The Launch of the Apple Macintosh (1984)





The psychology of human-computer interaction



The Psychology of Human-Computer Interaction

STUART K. CARD
THOMAS P. MORAN
ALLEN NEWELL



- Emerged from work done at Xerox PARC
- Card and Moran arrived at PARC in 1974
- Newell was a consultant to the project.
- This book provides basic psychology involving in the human sensory, cognition, and motor output.





HCI in the past

- Focused on personal behavior
- Focused on creating Software
- Research in HCI was done mainly for Desktop





HCI today

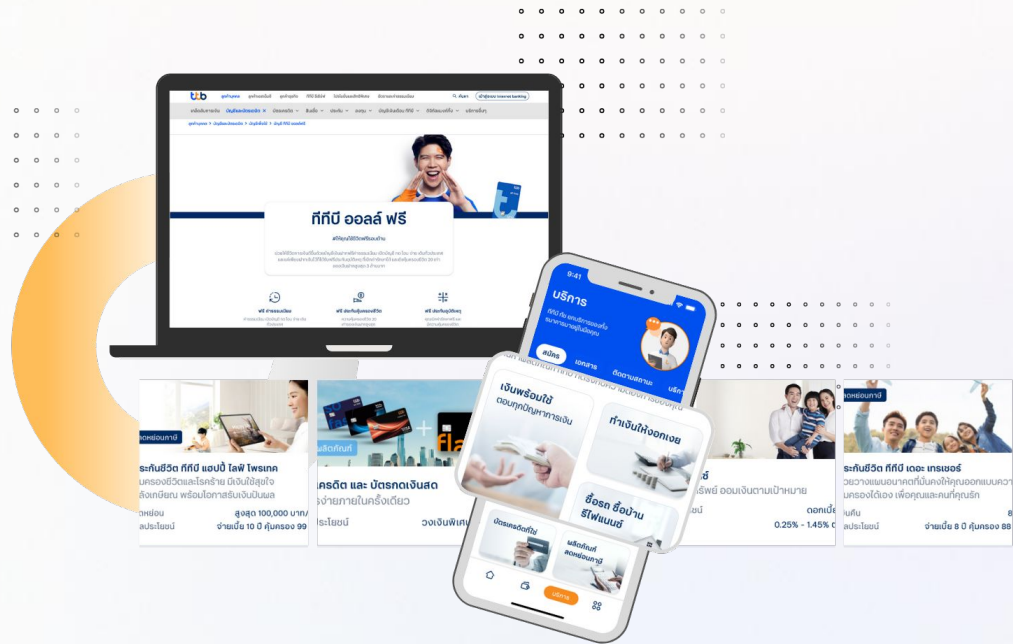
- Focused on social behavior
- Focused on users experience
- Research was done in different platforms



User experience



Interactive Technologies



The Humanoid AI Robots





Midterm - Keywords

UI

Usability

HCI

Agile
Vs
Waterfall

Software
Development Process

Metaphor

HTML/CSS



Human Perception



Web Architecture



Navigation





Post-test

What is this Module name?



Students choose an option

Pear Deck Interactive Slide
Do not remove this bar

Post-test

What is his name?



Students choose an option

Pear Deck Interactive Slide
Do not remove this bar

What does WYSIWYG stand for?



Post-test

Which study is not related to HCI ?



Students choose an option

Pear Deck Interactive Slide
Do not remove this bar

Post-test

What does API stand for?



Post-test

Which was the first computer designed specifically to be used with a mouse?



What does responsive design mean in web development?



Post-test

Which tool are we going to use to design
UI mockups in this course?

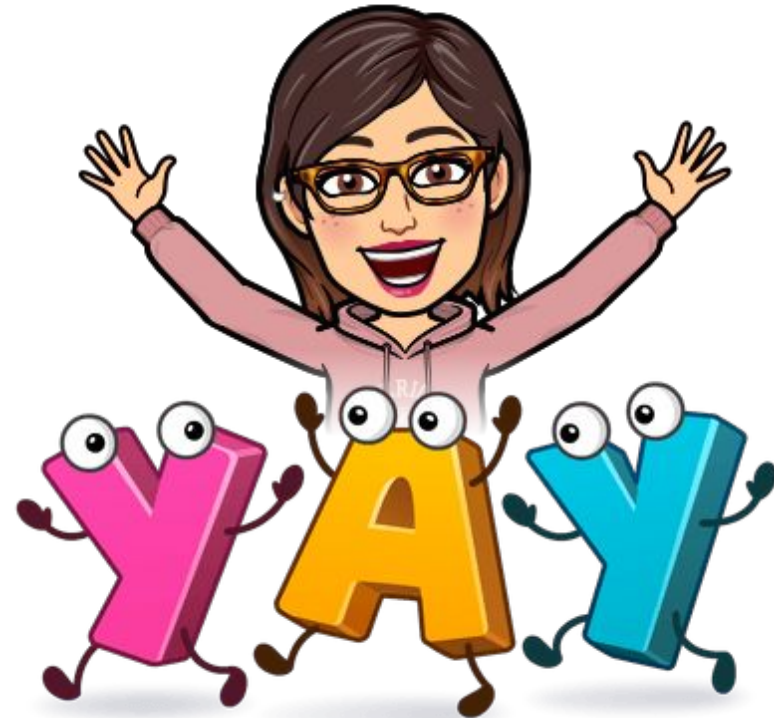


Students choose an option

Pear Deck Interactive Slide
Do not remove this bar

Post-test

What outcomes
do you expect
from this module?



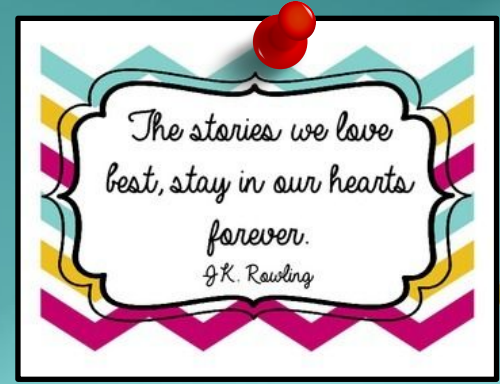
Students, write your response!



Checking your attendance



READING
gives us
someplace
to go when
we have to stay
where we
ARE



Contact me

You can contact me via :



Inthraporn.aranyanak@tudublin.ie

